



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to KTS NAGAMANICKAM CHETTIAR

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

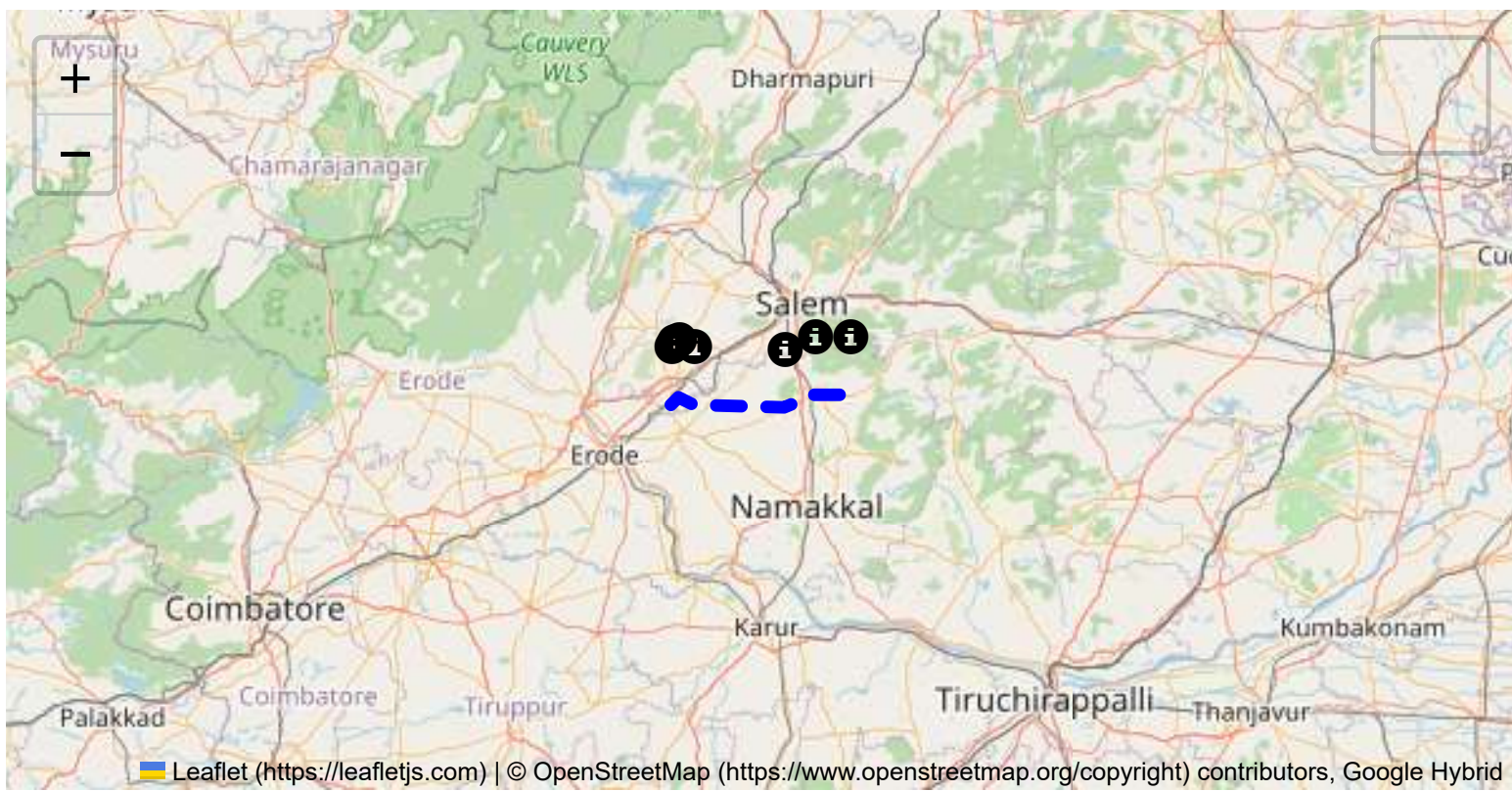
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

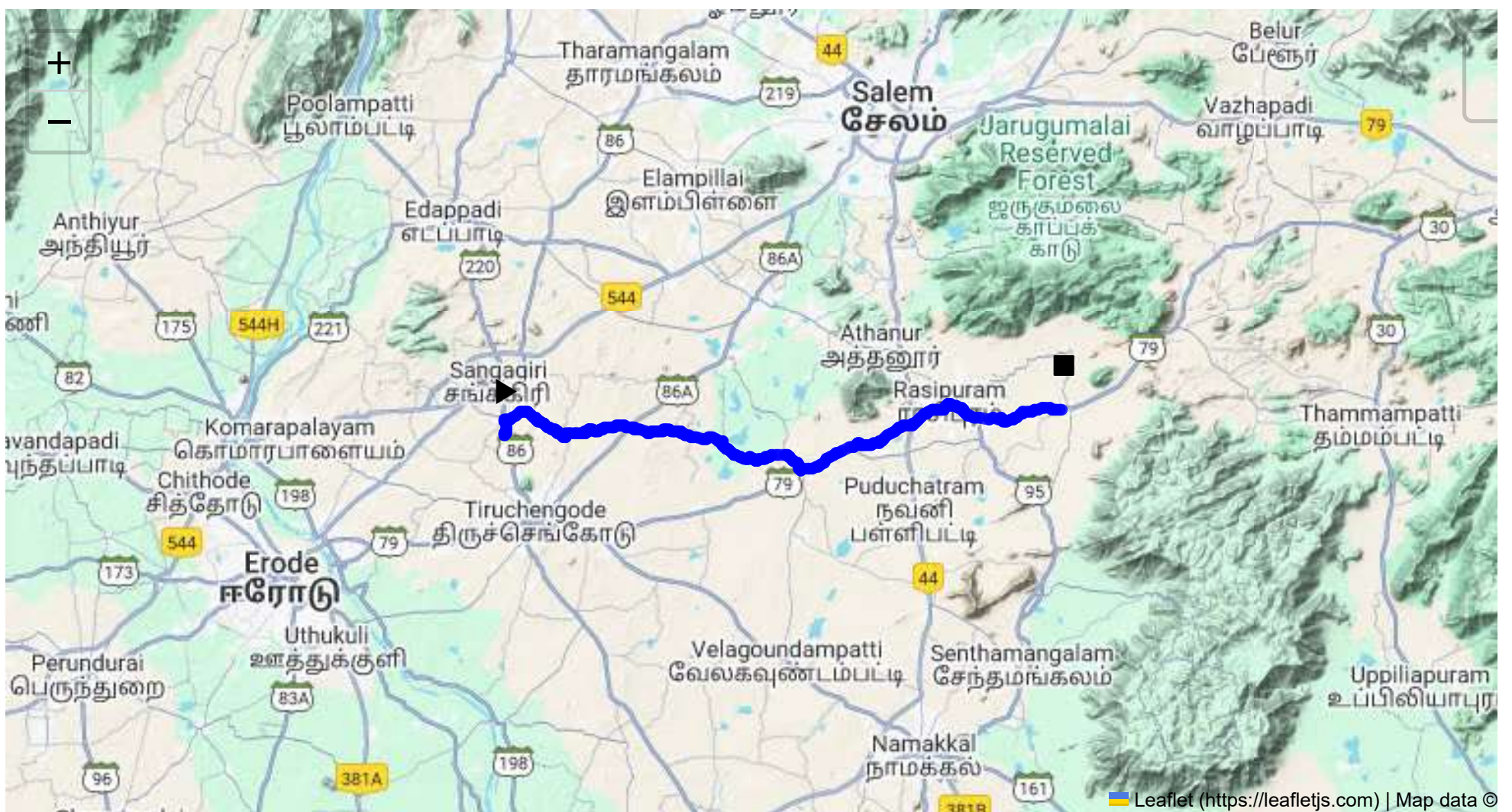
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 50.16 km
Estimated Duration: 1.4 hours
Adjusted Duration (Heavy Vehicle): 1.7 hours
Start: (11.4381, 77.8734)
End: (11.456174, 78.263237)



Welcome to the Journey Risk Management Study

Route Overview

The route from CVQF+23W, Sangagiri to F747+98X, Rasipuram via several intermediate points covers approximately 50.16 kilometers. Starting from Sangagiri, it moves through Sankari R.S, Kozhikkalnatham, and Ponkurichi before reaching Rasipuram. It includes transitioning from local roads to State Highway 79.

Typical Weather Conditions

Weather-Related Hazards

- **Monsoon Season (June to September):** Expect heavy rainfall, potentially causing waterlogging and reduced visibility.
- **Summer (March to May):** High temperatures might lead to road surface damage and heat stress for drivers.
- **Winter (November to February):** Generally mild, but fog can be a concern in early mornings.

Traffic Patterns

- **Peak Hours:** Typically, peak traffic is observed between 8-10 AM and 5-7 PM due to local commuter traffic.
- **Congestion-Prone Areas:** Key congestion points include market areas in Sangagiri and Ponkurichi, and junctions near the Rasipuram entry.

Road Quality and Infrastructure

- **Road Conditions:** Generally, SH 79 offers good quality roads, whereas local roads might have potholes and uneven surfaces, especially after rains.
- **Infrastructure:** Well-marked roads near urban areas, but rural stretches might lack adequate signage and lighting.

Alternative Routes for Emergencies

- Utilize State Highway 86 as an alternative which reconnects to SH 79 closer to Rasipuram, though it adds additional distance.

Local Regulations on Hazardous Material Transport

- **Permits Required:** Ensure to have the necessary permissions for transporting hazardous materials.
- **Time Restrictions:** Certain areas might have restrictions during peak hours to reduce congestion.

Historical Incidents

- **Major Incidents:** There have been few historical incidents involving heavy vehicles due to mechanical failures or driver error.
- **Safety Precautions:** Regular vehicle checks and driver training on local conditions are recommended.

Environmental Considerations

- **Sensitive Areas:** Pay attention to agricultural zones where farm vehicles might cross.
- **Preservation Zones:** Minimal environmental preservation zones; however, ensure no spillage in water crossing areas to prevent contamination.

Communication Coverage

- **Coverage Strength:** Mobile network coverage is generally good along SH 79 but expect potential dead zones on rural roads.
- **Ensure:** Carry additional communication devices or a reliable network provider.

Emergency Response Times

- **Urban Segments (near Sangagiri and Rasipuram):** 15-25 minutes for emergency services.
- **Rural Segments:** 30-45 minutes due to less immediate access.

Overall Summary of Risk Assessment

- **Primary Risks:** Include weather-related issues, traffic congestion, and potential road surface issues.
- **Mitigation Strategies:** Provide driver training on local conditions, have predefined communication protocols for emergencies, and establish contact with local emergency services.
- **Overall Evaluation:** The route is moderate risk, primarily due to weather and road quality. Proper precautions can effectively manage risks.

This comprehensive assessment equips the truck driver with necessary foresight into the journey, allowing safer transport of hazardous materials.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	Medium	11.43822, 77.87348	30 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Blind Spot	Blind Spot	11.44861, 77.87429	10 KM/Hr
4	Turn	Medium	11.45468, 77.88945	30 KM/Hr
5	Turn	Medium	11.44194, 77.90923	30 KM/Hr
6	Blind Spot	Blind Spot	11.43673, 77.91703	10 KM/Hr
7	Turn	High	11.43919, 77.91702	15 KM/Hr
8	Turn	Medium	11.43930, 77.91729	30 KM/Hr
9	Turn	Medium	11.43969, 77.92357	30 KM/Hr
10	Turn	Medium	11.43996, 77.92574	30 KM/Hr
11	Turn	Medium	11.43982, 77.93018	30 KM/Hr
12	Turn	Medium	11.44033, 77.93076	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
13	Turn	Medium	11.44071, 77.93211	30 KM/Hr
14	Turn	Medium	11.44096, 77.93223	30 KM/Hr
15	Turn	Medium	11.44135, 77.93220	30 KM/Hr
16	Turn	Medium	11.44175, 77.93249	30 KM/Hr
17	Turn	High	11.44248, 77.93247	15 KM/Hr
18	Turn	Medium	11.44289, 77.93420	30 KM/Hr
19	Turn	Medium	11.44094, 77.93824	30 KM/Hr
20	Turn	Medium	11.44547, 77.95309	30 KM/Hr
21	Turn	Medium	11.44370, 77.96020	30 KM/Hr
22	Turn	High	11.44302, 77.96480	15 KM/Hr
23	Turn	High	11.44243, 77.96500	15 KM/Hr
24	Turn	Medium	11.44011, 77.97521	30 KM/Hr
25	Turn	Medium	11.44158, 77.98455	30 KM/Hr
26	Turn	High	11.43528, 78.01534	15 KM/Hr
27	Turn	High	11.43720, 78.01774	15 KM/Hr
28	Turn	High	11.43427, 78.02658	15 KM/Hr
29	Turn	Medium	11.43135, 78.02685	30 KM/Hr
30	Turn	Medium	11.42818, 78.03138	30 KM/Hr
31	Turn	High	11.42324, 78.03848	15 KM/Hr
32	Turn	High	11.42366, 78.03889	15 KM/Hr
33	Turn	Medium	11.42222, 78.04112	30 KM/Hr
34	Turn	High	11.42216, 78.04283	15 KM/Hr
35	Turn	High	11.42261, 78.04307	15 KM/Hr
36	Turn	Medium	11.42260, 78.04326	30 KM/Hr
37	Turn	High	11.42151, 78.04462	15 KM/Hr
38	Turn	High	11.42157, 78.04478	15 KM/Hr
39	Blind Spot	Blind Spot	11.42318, 78.04539	10 KM/Hr
40	Turn	Medium	11.42120, 78.05015	30 KM/Hr
41	Turn	High	11.42195, 78.05572	15 KM/Hr
42	Turn	High	11.42259, 78.05580	15 KM/Hr
43	Turn	Medium	11.42448, 78.06061	30 KM/Hr
44	Turn	Medium	11.41967, 78.07833	30 KM/Hr
45	Turn	Medium	11.41669, 78.07955	30 KM/Hr
46	Blind Spot	Blind Spot	11.41453, 78.08062	10 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
47	Turn	Medium	11.41476, 78.08094	30 KM/Hr
48	Turn	Medium	11.43240, 78.12064	30 KM/Hr
49	Turn	Medium	11.43257, 78.12096	30 KM/Hr
50	Turn	Medium	11.44497, 78.15275	30 KM/Hr
51	Turn	Medium	11.45586, 78.18045	30 KM/Hr
52	Turn	High	11.45602, 78.18078	15 KM/Hr
53	Turn	Medium	11.45739, 78.18121	30 KM/Hr
54	Turn	Medium	11.46002, 78.18424	30 KM/Hr
55	Blind Spot	Blind Spot	11.46035, 78.18534	10 KM/Hr
56	Turn	High	11.45904, 78.18556	15 KM/Hr
57	Turn	Medium	11.45864, 78.18755	30 KM/Hr
58	Turn	Medium	11.45132, 78.19979	30 KM/Hr
59	Turn	Medium	11.44994, 78.22138	30 KM/Hr
60	Turn	Medium	11.44919, 78.22291	30 KM/Hr
61	Turn	High	11.44806, 78.22308	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
7	hospital	Sudha Hospital	11.4555773, 78.1760125	30 km/h	Medium
8	hospital	Government Hospital Blood Bank, Rasipuram	11.4612082, 78.1849249	30 km/h	Medium
9	hospital	M.R. Hospital, Namakkal	11.4613462, 78.1861702	30 km/h	Medium
10	hospital	dr.thangavel dch hospital	11.4619955, 78.1849181	30 km/h	Medium
11	police	Police Station Rasipuram	11.4617825, 78.1847411	30 km/h	Medium
12	hospital	Government Hospital, Rasipuram	11.4612987, 78.1851369	30 km/h	Medium
13	hospital	Rasi Hospital	11.4618368, 78.1855695	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
14	hospital	surya hospital	11.4622163, 78.1840544	30 km/h	Medium
15	hospital	R K Eye Care Center	11.4589079, 78.1913596	30 km/h	Medium
16	hospital	Manickkam Hospital	11.4568944, 78.1926061	30 km/h	Medium
18	hospital	Snathi Hospital	11.4564245, 78.1932485	30 km/h	Medium
19	hospital	Sugam Hospital,Rasipuram	11.4553007, 78.1936221	30 km/h	Medium
20	hospital	Government Hospital, Namagiripettai	11.4569484, 78.265263	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	Sri Vinayaga Matriculation Higher Secondary School	11.4233682, 78.0736517	30 km/h	Medium
1	marketplace	Farmer Market	11.4146849, 78.0805463	30 km/h	Medium
2	school	Government Higher Secondary School	11.4150045, 78.0838172	30 km/h	Medium
3	school	Vidya madir school	11.4356335, 78.1381408	30 km/h	Medium
4	college	Thiruvalluvar Govt arts college	11.4449352, 78.1595562	30 km/h	Medium
5	school	Venkateswara School	11.4468258, 78.1595761	30 km/h	Medium
6	school	Venkateswara Middle school	11.4472063, 78.1596191	30 km/h	Medium
17	school	Govt. Hr. Sec. School	11.4585324, 78.1933666	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44861, 77.87429



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45468, 77.88945



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44194, 77.90923



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.43673, 77.91703



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.43919, 77.91702



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.43930, 77.91729



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43969, 77.92357



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43996, 77.92574



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43982, 77.93018



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44033, 77.93076



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44071, 77.93211



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44096, 77.93223



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44135, 77.93220



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44175, 77.93249



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44248, 77.93247



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44289, 77.93420



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44094, 77.93824



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44547, 77.95309



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44370, 77.96020



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44302, 77.96480



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44243, 77.96500



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44011, 77.97521



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44158, 77.98455



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.43528, 78.01534



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.43720, 78.01774



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.43427, 78.02658



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43135, 78.02685



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.42818, 78.03138



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.42324, 78.03848



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.42366, 78.03889



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.42222, 78.04112



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.42216, 78.04283



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.42261, 78.04307



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.42260, 78.04326



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.42151, 78.04462



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.42157, 78.04478



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.42318, 78.04539



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.42120, 78.05015



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.42195, 78.05572



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.42259, 78.05580



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.42448, 78.06061



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41967, 78.07833



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41669, 78.07955



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.41453, 78.08062



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41476, 78.08094



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.43240, 78.12064



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.43257, 78.12096



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44497, 78.15275



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45586, 78.18045



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45602, 78.18078



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45739, 78.18121



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46002, 78.18424



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.46035, 78.18534



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45904, 78.18556



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45864, 78.18755



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45132, 78.19979



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44994, 78.22138



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44919, 78.22291



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44806, 78.22308

